

Complex Analysis

5.9 Real integral using contour integration

$z = re^{i\theta}$ (equation of a circle of radius r)

$$z = e^{i\theta} \quad (r = 1) \Rightarrow dz = ie^{i\theta} d\theta \Rightarrow dz = iz d\theta \Rightarrow d\theta = \frac{dz}{iz}$$

When $z = e^{i\theta}$ & $\frac{1}{z} = e^{-i\theta}$ Then

$$\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right) \quad \& \quad \cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

Example: Find the residue of the following functions:

$$(a) \quad f(\theta) = \int_0^{2\pi} \sin \theta d\theta$$

$$\text{Solution: } \int_0^{2\pi} \sin \theta d\theta = \oint_c \frac{1}{2i} \left[z - \frac{1}{z} \right] \frac{dz}{iz} = -\frac{1}{2} \oint_c \left[1 - \frac{1}{z^2} \right] dz \quad c: |z|=1$$

b_1 (coefficient of $\frac{1}{z}$) = 0 so Residue $I = 0$

$$(b) \quad f(\theta) = \int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta}$$

Solution: $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} = \oint_c \frac{dz}{iz \left[\sqrt{2} - \frac{1}{2} \left(\frac{1}{z} + z \right) \right]}$, $c: |z|=1$

$$f(z) = \oint_c \frac{dz/iz}{\sqrt{2} - \frac{1}{2} \left(z + \frac{1}{z} \right)} = \oint_c \frac{dz/iz}{(2\sqrt{2}z - z^2 - 1)/2z}$$

$$= \frac{2}{i} \oint_c \frac{dz}{z^2 - 2\sqrt{2}z + 1}, \quad z^2 - 2\sqrt{2}z + 1 = 0 \Rightarrow z = \sqrt{2} \pm 1$$

$$= -\frac{2}{i} \oint \frac{dz}{(z-\alpha)(z-\beta)}, \quad \alpha = \sqrt{2} + 1 \text{ \& \; } \beta = \sqrt{2} - 1$$

$$\oint f(z)dz = -\frac{2}{i} \times 2\pi i \times \frac{1}{\beta - \alpha} = -2 \times 2\pi \times \left(\frac{-1}{2} \right) = 2\pi$$

$$(c) \quad f(\theta) = \int_0^{2\pi} \frac{1 + \sin \theta}{3 + \cos \theta} d\theta$$

Solution: $\int_0^{2\pi} \frac{1 + \sin \theta}{3 + \cos \theta} d\theta = \oint_c \frac{1}{3 + \frac{1}{2} \left[z + \frac{1}{z} \right]} \frac{dz}{iz} + \text{Im} \oint \frac{z}{3 + \frac{1}{2} \left(z + \frac{1}{z} \right)} \frac{dz}{iz} \Rightarrow z = 1.e^{i\theta}, \text{Im } z = \sin \theta$

$$= \oint_c \frac{1}{\left[3 + \frac{z}{2} + \frac{1}{2z} \right]} \frac{dz}{iz} + \text{Im} \oint \left[\frac{z}{3 + \frac{z}{2} + \frac{1}{2z}} \right] \frac{dz}{iz}$$

$$= \oint \left[\frac{2z}{6z + z^2 + 1} \right] \frac{dz}{iz} + \text{Im} \oint \left[\frac{2z^2}{6z + z^2 + 1} \right] \frac{dz}{iz}$$

$$z^2 + 6z + 1 = 0 \Rightarrow z = -3 \pm 2\sqrt{2} \Rightarrow \alpha = -3 + 2\sqrt{2}, \quad \beta = -3 - 2\sqrt{2}$$

$$\oint f(z)dz = \frac{2}{i} \times 2\pi i \times \frac{1}{\alpha - \beta} + \text{Im} \frac{2}{i} \times 2\pi i \times \frac{\alpha}{\alpha - \beta} = \frac{4\pi}{\alpha - \beta} + 0 = \frac{4\pi}{4\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

$$(d) \quad f(\theta) = \int_0^{2\pi} \frac{\sin^2 \theta}{5 - 4\cos \theta} d\theta$$

Solution: $\int_0^{2\pi} \frac{\sin^2 \theta}{5-4\cos \theta} d\theta = -\frac{1}{2} \int_0^{2\pi} \frac{(1-\cos 2\theta)}{(5-4\cos \theta)} d\theta$

Let $z = e^{i\theta}$ and $z^2 = e^{2i\theta} \Rightarrow \operatorname{Re} z^2 = \cos 2\theta \Rightarrow \frac{1-\cos 2\theta}{2} \Rightarrow \operatorname{Re} \left[\frac{1-e^{2i\theta}}{2} \right]$

$$= \operatorname{Re} \oint_c \frac{1-z^2}{2 \left(5 - \frac{4}{2} \left(z + \frac{1}{z} \right) \right) iz} dz = \operatorname{Re} \oint_c \frac{1-z^2}{2 \left(5 - 2z - \frac{2}{z} \right) iz} dz$$

$$= \operatorname{Re} \oint_c \frac{1-z^2}{(10z-4z^2-4)} = \operatorname{Re} \oint_c \frac{1-z^2}{2i(5z-2z^2-2)}$$

$$= \operatorname{Re} \left[-\frac{1}{2i} \oint_c \frac{(1-z^2)}{2z^2-5z+2} dz \right] = -\frac{1}{4i} \oint_c \frac{(1-z^2)}{z^2 - \frac{5}{2}z + 1} dz$$

$$z^2 - \frac{5}{2}z + 1 = 0 \Rightarrow z = \frac{5 \pm 3}{4} = 2, \frac{1}{2} \Rightarrow \alpha = 2, \beta = \frac{1}{2}$$

$$R = \operatorname{Re} \left(\frac{-1}{4i} \times 2\pi i \times \left(\frac{1-\frac{1}{4}}{\left(\frac{1}{2} - 2 \right)} \right) \right) = \frac{1}{2} \pi \times \frac{3/4}{3/2} = \frac{\pi}{2} \times \frac{3}{4} \times \frac{2}{3} = \frac{\pi}{4}$$

(e) $f(\theta) = \int_0^\pi \frac{1}{5-4\cos \theta} d\theta$

Solution: $f(\theta) = \int_0^\pi \frac{1}{5-4\cos \theta} d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1}{5-4\cos \theta} d\theta = \frac{1}{2} \oint_c \frac{dz / iz}{5-2 \left(z + \frac{1}{z} \right)}$

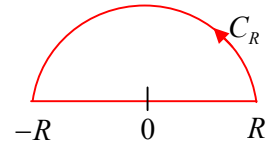
$$= \frac{1}{2} \oint_c \frac{dz}{\left(5 - 2z - \frac{2}{z} \right) iz} = \frac{1}{2i} \oint_c \frac{1}{5-2z^2-2} = -\frac{1}{4i} \oint_c \frac{dz}{z^2 - \frac{5}{2}z + 1}$$

$$z = \frac{5 \pm 3}{4}, \alpha = 2, \beta = \frac{1}{2}$$

$$z = -\frac{1}{4i} \times 2\pi i \times \left(\frac{1}{\left(\frac{1}{2} - 2 \right)} \right) = \frac{\pi}{3}$$

Case 3. $\int_{-\infty}^{\infty} f(x) dx$

When R goes to infinity, the whole of the upper half-plane is covered. If the coefficient of i is positive in the pole, the pole is inside of the contour.



$$\oint_C f(z) dz = 2\pi i \sum R$$

$$\int_{cR} f(z) dz + \int_{-R}^R f(x) dx = 2\pi i \sum R$$

$$\int_{cR} f(z) dz + \int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum R$$

If $R \rightarrow \infty$, Then $\lim_{R \rightarrow \infty} \int_{cR} f(z) dz + \int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum R$

Put $z = re^{i\phi}$, Then solve limit $\lim_{R \rightarrow \infty} \int_{cR} f(z) dz = 0$

If poles lie inside the contour then $\Rightarrow \int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum R$

And If the poles lie on the contour then $\Rightarrow \int_{-\infty}^{\infty} f(x) dx = \pi i \sum R$.

Example: $f(x) = \int_{-\infty}^{\infty} \frac{\sin x}{x} dx$

Solution: $f(z) = \frac{e^{iz}}{z} \Rightarrow \text{Im} \oint_C \frac{e^{iz}}{z} dz$

$z = 0$, pole which lies on the real plane. So $\oint_C f(z) dz = \pi i \sum R$

$\sum R = e^{i \cdot 0} = e^0 = 1 \Rightarrow \text{Im}(\pi i \times 1) = \pi \Rightarrow \oint_C f(z) dz = \pi$

Example: $f(x) = \int_{-\infty}^{\infty} \frac{\sin x}{x^3} dx$

Solution: $f(z) = \frac{e^{iz}}{z^3} \Rightarrow \frac{1}{z^3} \left(1 + iz + \frac{(iz)^2}{2!} + \frac{(iz)^3}{3!} + \dots \right)$

$b_1 = -\frac{1}{2}$ (Coefficient of $\frac{1}{z}$), Residue $I = \text{Im} \left[-\frac{1}{2} \times \pi i \right] = -\frac{\pi i}{2}$, $I = -\frac{\pi}{2}$

Example: $f(x) = \int_{-\infty}^{\infty} \frac{\sin 2x}{x^3} dx$

Solution: $f(z) = \frac{e^{2iz}}{z^3}$

$$\oint_c f(z) dz = \pi i \sum R = \frac{\pi i}{2!} \frac{d^2}{dz^2} e^{2iz} = -2\pi i$$

$$\text{Im} \oint_c f(z) dz = -2\pi$$

Example: $f(x) = \int_0^{\infty} \frac{1}{1+x^4} dx$

Solution: $f(z) = \frac{1}{z^4 + 1}$, $\phi(z) = 1$ & $\psi(z) = z^4 + 1$

$$\Rightarrow z^4 + 1 = 0, \quad z^4 = -1 = e^{i(2n+1)\pi}, \quad z = e^{i(2n+1)\pi/4}$$

$$z_0 = e^{i\pi/4} (+ve), z_1 = e^{i3\pi/4} (+ve), z_2 = e^{i5\pi/4} (-ve), z_3 = e^{i7\pi/4} (-ve)$$

$$z_0 = \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}, z_1 = -\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$$

Residue at z_0 : $\lim_{z \rightarrow z_0} \frac{1}{4z^3} = \frac{1}{4} e^{-3i\pi/4}$

Residue at z_1 : $\lim_{z \rightarrow z_1} \frac{1}{4z^3} = \frac{1}{4} e^{-i9\pi/4}$

$$\text{Residue } I = \frac{2\pi i}{4} [e^{-3i\pi/4} + e^{-i9\pi/4}] = \frac{\pi i}{2} \left[\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4} + \cos \frac{9\pi}{4} - i \sin \frac{9\pi}{4} \right]$$

$$I = \frac{\pi i}{2} \left[\frac{-1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right] = \frac{\pi}{\sqrt{2}}$$

2. Find the principal value of $\int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)}$

Solution: $\int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)} = \int_0^{\infty} \frac{dz}{(z^2 - 3z + 2)(z^2 + 1)}$

$$f(z) = \frac{dz}{(z-1)(z-2)(z+i)(z-i)}$$

$z = -i$ does not lie on contour.

$z = 1, 2$ lie on the real axis so during residue we will multiply the term by πi rather than $2\pi i$.

$$z = i, \quad \lim_{z \rightarrow i} \frac{dz}{(z-1)(z-2)(z+i)} = \frac{1}{(i-1)(i-2)(2i)} \quad (1)$$

$$z = 1, \quad \lim_{z \rightarrow 1} \frac{dz}{(z-i)(z-2)(z+i)} = \frac{1}{(1-i)(-1)(1+i)} = \frac{1}{(i-1)(i+1)} = \frac{-1}{2} \quad (2)$$

$$z = 2, \quad \lim_{z \rightarrow 2} \frac{dz}{(z-i)(z-1)(z+i)} = \frac{1}{(1)(2+i)(2-i)} = \frac{1}{4+1} = \frac{1}{5} \quad (3)$$

$$\text{Residue } I = \pi i \left[\frac{1}{5} - \frac{1}{2} \right] + 2\pi i \left[\frac{1}{(2i)(i-1)(i-2)} \right]$$

$$I = -\frac{3\pi i}{10} + \pi \left[\frac{1}{-1-2i-i+2} \right] = -\frac{3\pi i}{10} + \pi \left[\frac{1}{1-3i} \times \frac{1+3i}{1+3i} \right]$$

$$I = -\frac{3\pi i}{10} + \pi \frac{(1+3i)}{1+9} = -\frac{3\pi i}{10} + \frac{\pi}{10} + \frac{3\pi i}{10}$$

$$\Rightarrow I = \frac{\pi}{10}$$

Example: Given $\int_0^\infty \frac{dx}{(y^2+x^2)} = \frac{\pi}{2y}$, find the value of $\int_0^\infty \frac{dx}{(y^2+x^2)^2} = ?$

Solution: $\int_0^\infty \frac{dx}{(y^2+x^2)^2} = \frac{1}{2} \int_{-\infty}^\infty \frac{dx}{(y+ix)^2 (y-ix)^2}$

$$\frac{\partial}{\partial y} \int_0^\infty \frac{dx}{(y^2+x^2)} = \frac{\partial}{\partial y} \left(\frac{\pi}{2y} \right) \Rightarrow \int_0^\infty \frac{-dx \cdot 2y}{(y^2+x^2)^2} = \frac{-\pi}{2y^2}$$

$$\Rightarrow \int_0^\infty \frac{dx}{(y^2+x^2)^2} = \frac{\pi}{4y^3}$$

Example: $f(x) = \int_0^\infty \frac{dx}{(x^4+4)}$

Solution: $\int_0^\infty \frac{dx}{(x^4+4)} = \frac{1}{2} \int_{-\infty}^\infty \frac{dx}{(x^4+4)}, \quad f(z) = \frac{1}{z^4+4}$

$$z^4 + 4 = 0 \Rightarrow z^4 = -4 = 4(-1) = 4e^{i(2n+1)\pi}$$

$$z_0 = \sqrt{2}e^{i\pi/4}, z_1 = \sqrt{2}e^{i3\pi/4}, z_2 = \sqrt{2}e^{i5\pi/4}, z_3 = \sqrt{2}e^{i7\pi/4}$$

$$f(z) = \frac{\phi(z)}{\psi(z)} \Rightarrow f'(z) = \frac{\phi'(z)}{\psi'(z)} = \frac{1}{4z^3}$$

$$\lim_{z \rightarrow \sqrt{2}e^{i\pi/4}} \frac{1}{4z^3} = \frac{1}{42\sqrt{2}} \times e^{-i\pi 3/4}$$

$$\lim_{z \rightarrow \sqrt{2}e^{3i\pi/4}} \frac{1}{4z^3} = \frac{1}{42\sqrt{2}} \times e^{-i9\pi/4}$$

$$\text{Then } \int_{-\infty}^{\infty} \frac{dx}{x^4} = 2\pi i \sum R = \frac{2\pi i}{8\sqrt{2}} [e^{-i3\pi/4} + e^{-i9\pi/4}]$$

$$= \frac{2\pi i}{8\sqrt{2}} \left[-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \right] = \frac{\pi}{4} \Rightarrow \int_0^{\infty} \frac{dx}{x^4 + 4} = \frac{1}{2} \times \frac{\pi}{4} = \frac{\pi}{8}$$